

VayuSetu

Find the pollution hotspot on your street,
prove it, and get it fixed.

Citizen photos, ground stations and satellites, verified by Gemini and routed to the agency that can clear it.

7

BRICS capitals live

77

ground stations streaming

13

languages

4

Gemini jobs, all grounded

Live: cleanair-backend--cleanair-clear-streets.asia-southeast1.hosted.app

THE PROBLEM

A city-wide AQI hides the air people breathe

99%

of people live where WHO air-quality guidelines are not met

4.2 M

premature deaths from outdoor air pollution in 2019

89%

of those deaths are in low- and middle-income countries



Street-level sources go unseen

A garbage fire or a construction site kilometres from the nearest monitor vanishes into the city average.



Pollution crosses borders

Crop burning and haze drift across state and national lines faster than one agency can track.



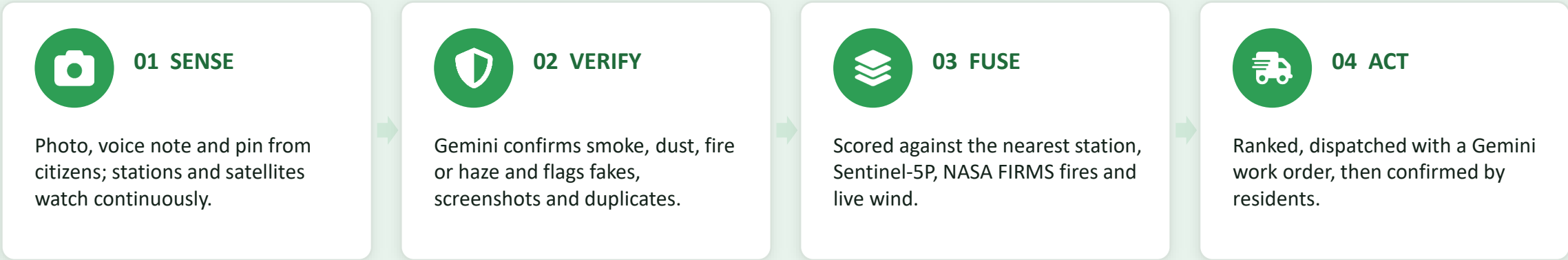
Data rarely reaches a crew

Dashboards publish numbers, but nobody is told which street to inspect or which department owns the fix.



One monitor, many invisible sources, and smoke that does not stop at the border.

From a citizen's photo to a verified, dispatched fix

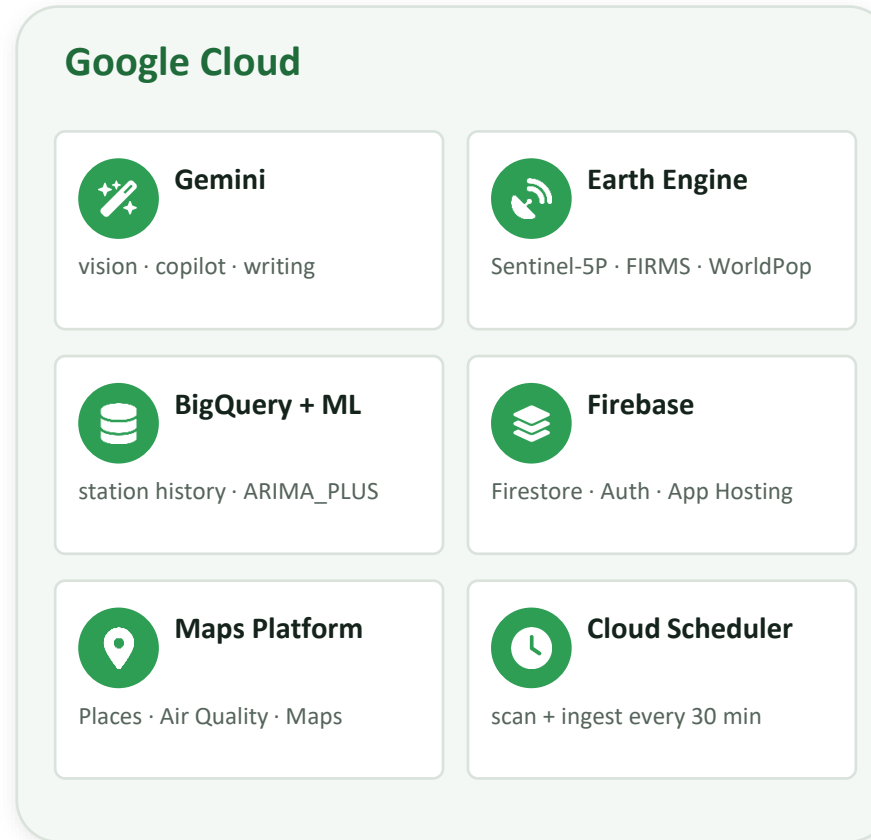
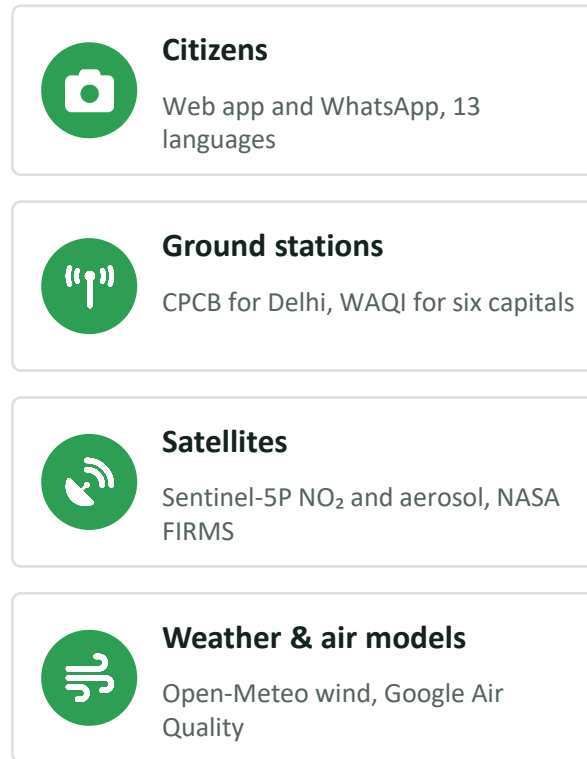


Not another AQI dashboard

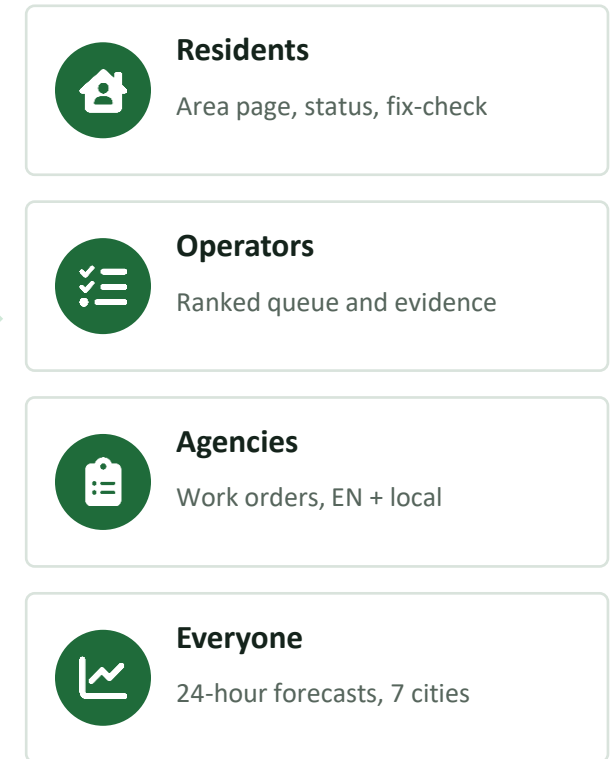
	TYPICAL AQI DASHBOARD	VAYUSETU
Scale	One city-wide average	Street-level H3 cells of ~0.7 km ²
Evidence	A single sensor feed	Citizens + stations + Sentinel-5P + FIRMS + wind
Trust	Reports taken at face value	Gemini, EXIF and duplicate checks; independent agreement
Outcome	Displays a number	Ranked queue, routed work order, residents confirm the fix

One hexagon grid ties every source together

INPUTS



OUTPUTS



Key decision: every report, station reading and satellite pixel is indexed to the same H3 hexagon ($\sim 0.7 \text{ km}^2$), so different evidence can be compared in the same place.

Gemini does four jobs and shows its evidence



Photo check

Labels smoke, dust, fire or haze with severity, and flags screenshots, edits and stock images.

Vision + JSON output



Operations copilot

Answers operators by calling live tools: incidents, stations, wind, fires, forecasts, upwind sources.

Function calling · 7 tools



Work orders

Drafts the order for the right agency, citing the evidence, in English and the city's language. For a repeat hotspot, asks for action at the source.

Structured output



Resident summaries

Explains what's happening on each street in the reader's language, with read-aloud.

13 languages · cached

Guardrails

Facts in, facts cited

Prompts carry only recorded evidence; outputs list what they used.

Never blank, never faked

Flash-Lite fallback; if both fail, the page says so.

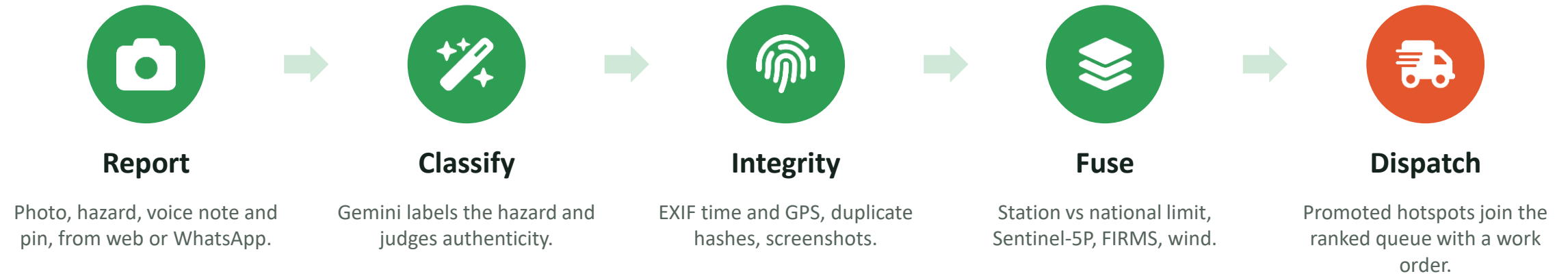
Cached per cell

Summaries regenerate only when the evidence changes.

Health advice

Comes from Google's Air Quality API, not from Gemini.

No alert without independent evidence

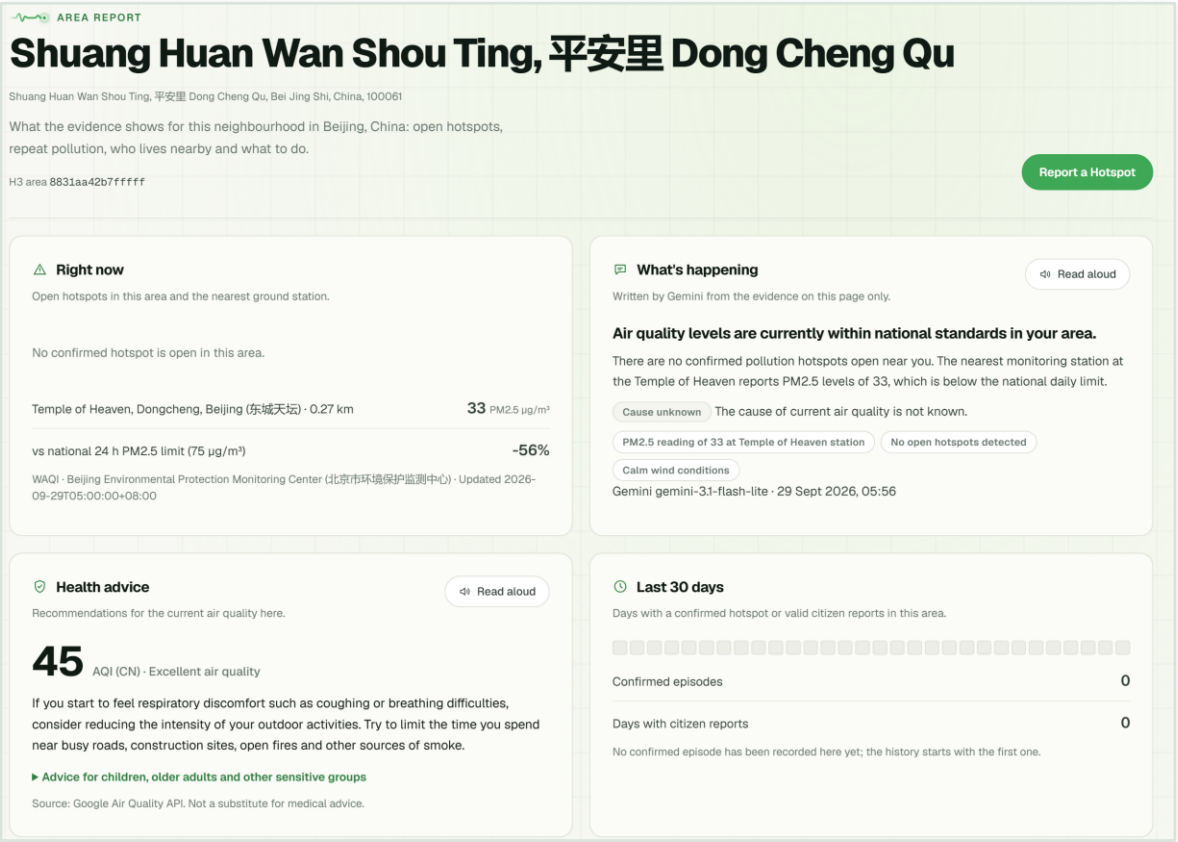
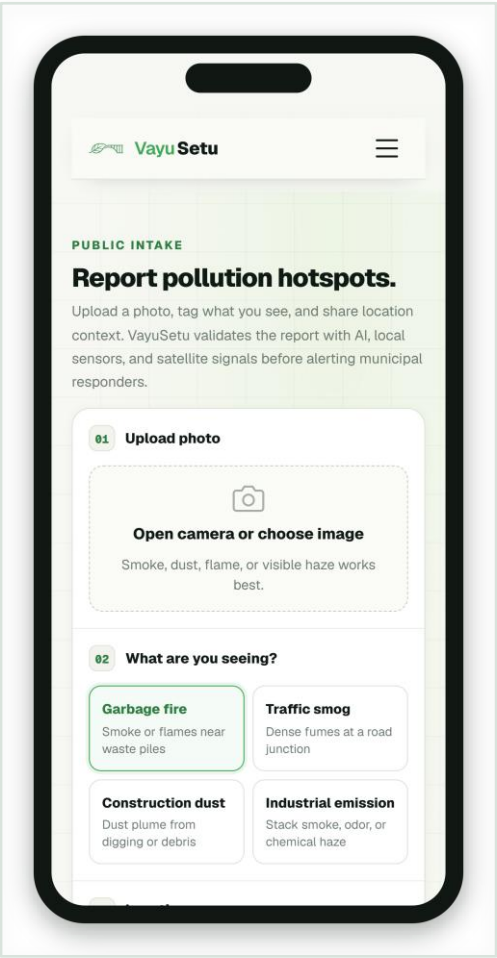


Promotion tiers, strongest first



46 of 48 incidents on record were raised automatically: a scan checks every station and Sentinel-5P every 30 minutes, with no photo needed. Flagged reports stay visible but never count toward promotion.

Report in seconds, see what's on your street



Live: Temple of Heaven area, Beijing. Nearest station 0.27 km away; ≈71,838 residents within 1 km (WorldPop 2020).



Photo, voice, pin
13 languages; Gemini checks the photo first.



A page for every street
Station, Gemini summary, health advice, history.

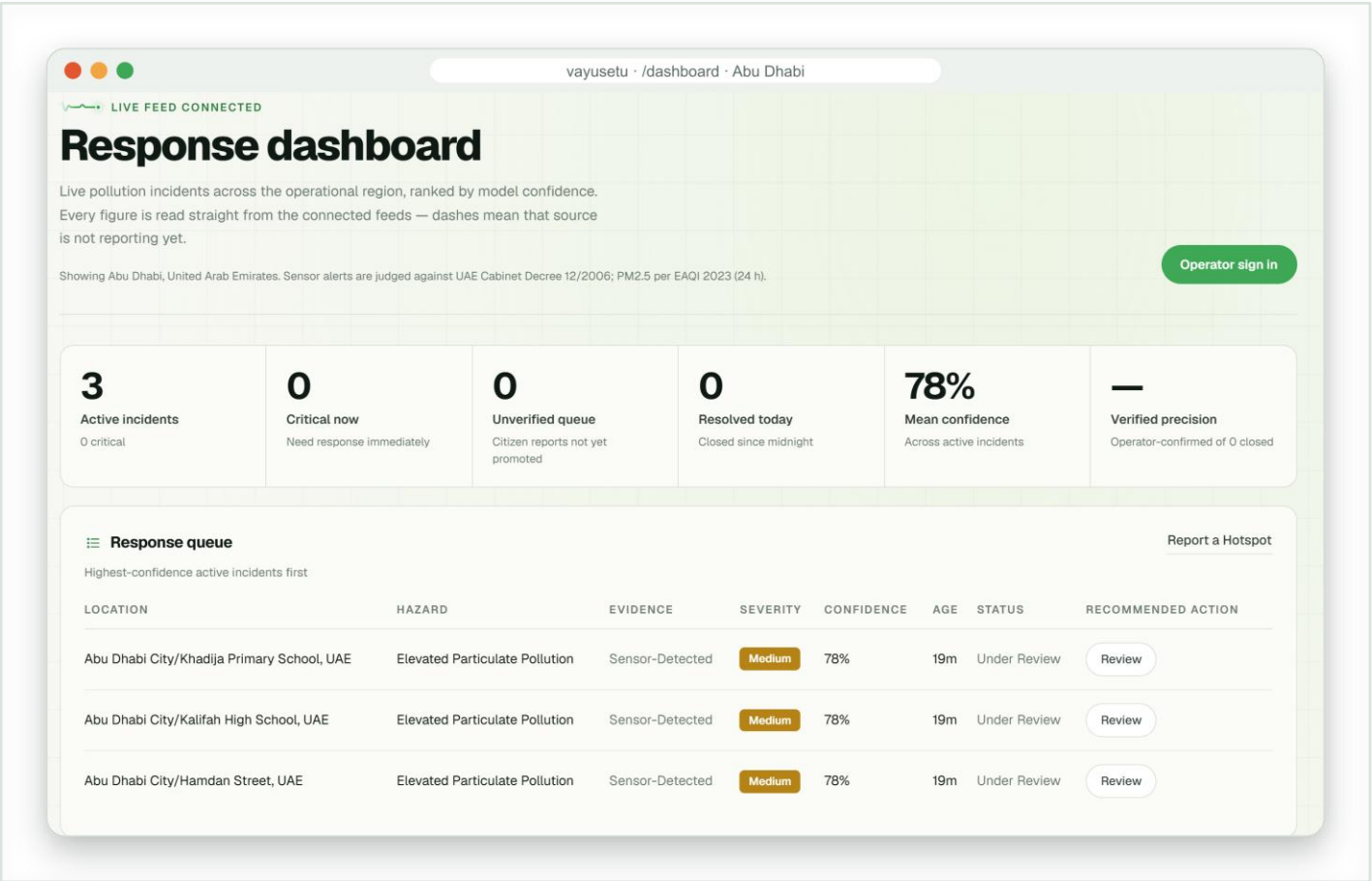


Who's affected
WorldPop residents, schools and hospitals within 1 km.

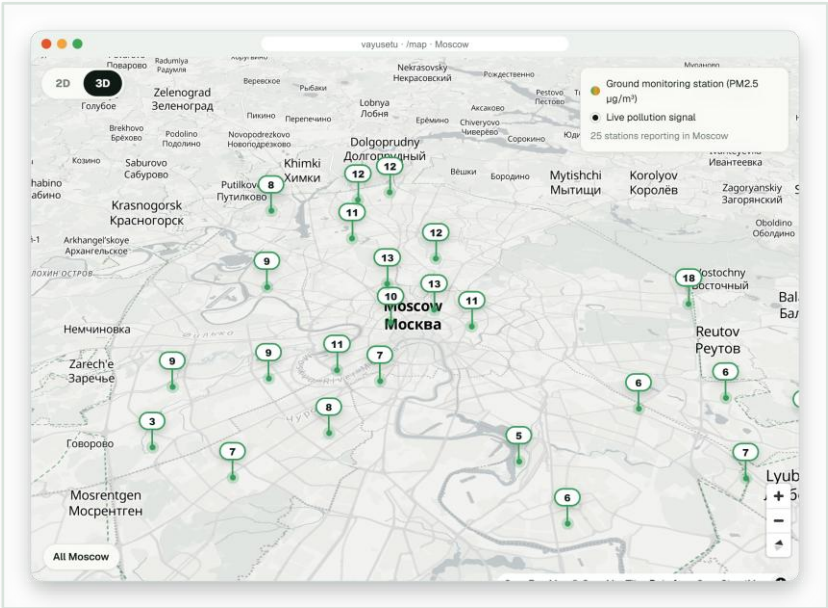


"Is it fixed?"
Reporters confirm or dispute a fix for 72 h.

Operators see what to fix first, and why



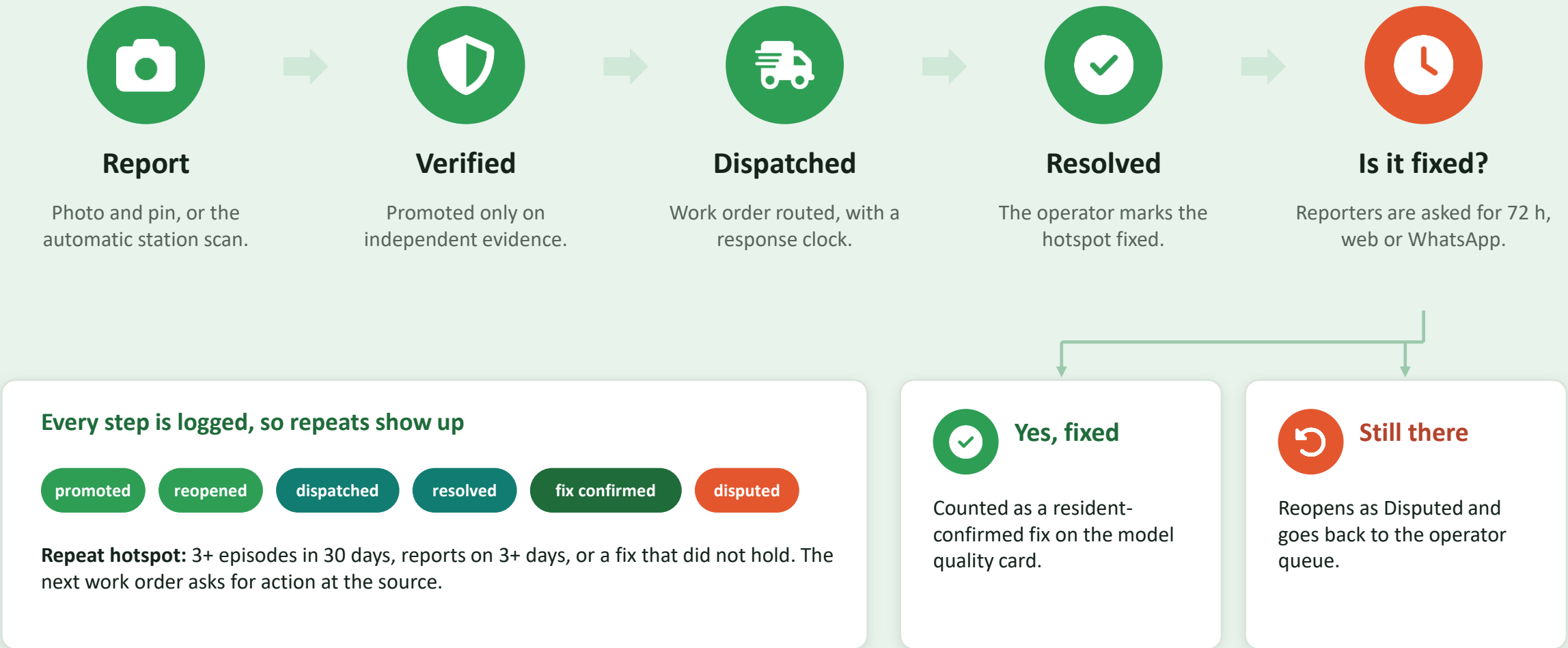
Abu Dhabi, live: particulate hotspots raised automatically from official EAD stations, judged against the UAE limit.



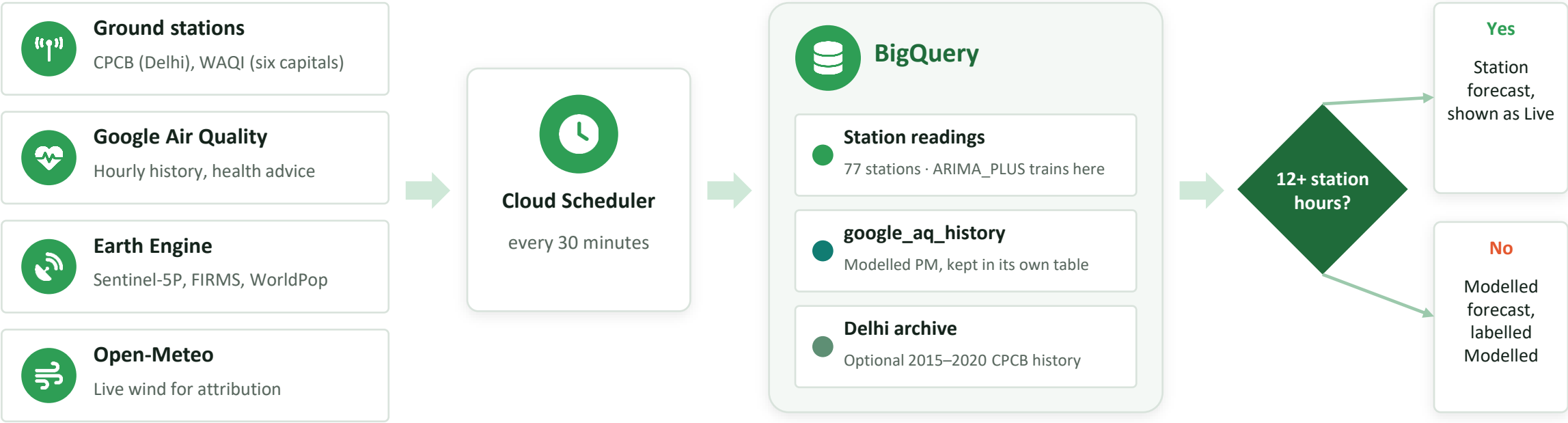
Moscow, 3D view: official stations reporting live.

- Ranked queue**
Evidence tier first; repeat, overdue and disputed chips.
- Traceable evidence**
Station vs national limit, satellite window, reports.
- One-click action**
Draft a work order, dispatch, resolve; reporters are told.

A fix counts only when residents confirm it



From raw feeds to an honest 24-hour outlook



7 / 7

capitals with a 24-hour forecast

77

ground stations streaming into BigQuery

50

forecast zones with a week of modelled history

9,191

modelled hourly rows stored

Real station readings always win; modelled history is Google's 500 m estimate and never appears as live. Counts as of 29 Sep 2026.

Seven capitals, seven legal limits, one engine



77 ground stations streaming every 30 min

46 of 48 hotspots raised automatically

≈71,838 residents counted near one Beijing landmark

New Delhi



60 $\mu\text{g}/\text{m}^3$
PM2.5 24 h limit

Beijing



75 $\mu\text{g}/\text{m}^3$
PM2.5 24 h limit

Moscow



35 $\mu\text{g}/\text{m}^3$
PM2.5 24 h limit

Pretoria



40 $\mu\text{g}/\text{m}^3$
PM2.5 24 h limit

Abu Dhabi



60 $\mu\text{g}/\text{m}^3$
PM2.5 24 h limit

Jakarta



55 $\mu\text{g}/\text{m}^3$
PM2.5 24 h limit

Brasília



50 $\mu\text{g}/\text{m}^3$
PM2.5 24 h limit

WHAT'S NEXT

Clean air, verified street by street, across BRICS.

Next 90 days

01

Pilot with one municipal team

Real dispatches and their response targets, instead of our placeholders.

02

Alerts near you

A WhatsApp message when a hotspot is confirmed within 1 km.

03

Proof of the fix

An after-photo from the crew, checked by Gemini against the report.

Built on Google Cloud

Gemini

Earth Engine

BigQuery ML

Firebase

Maps Platform

Air Quality API

Cloud Scheduler

Citizens

report in their language and see what happened

Operators

act on ranked, evidence-backed hotspots

Nations

share one engine and keep their own rules

Thank you